offer AI-powered screening packages to predict the likelihood of different ailments (see *Where Do You Get Tested*). Most algorithms rely on pre-fed patient data set and medical literature to make the predictions. In the case of type-2 diabetes, for example, obesity is found to increase its risk by six times, according to a 2020 research published in the medical journal *Diabetologia*. Another 2007 study in the same journal found variations in the TCF7L2 gene to be strongly linked to type-2 diabetes in the Indian population. Research has also found that poor sleep (a 2012 study published in the medical journal *BMJ*) and family history (a 2017 study conducted in Tamil Nadu and published in the *International Journal of Community Medicine and Public Health*) add to the diabetes risk. Thus, if a patient is found to be obese, carrying a gene mutation, having poor sleep and a family history of diabetes, then the algorithm will predict a higher than average chance of developing type-2 diabetes.

Another condition is that of familial hypercholesterolemia, a genetic disorder. It is caused by a defect in chromosome 19, which makes the body unable to remove the low-density lipoprotein (LDL), or bad cholesterol, from the blood. If a patient is found to have this gene, along with other risk factors like obesity, family history and a smoking habit, the AI algorithm will predict a high likelihood of heart attack, unless steps are taken to prevent it—even if, at the present, the person has no blockages in the heart.

Indeed, almost all the latest diagnostics for predicting disease revolves around AI technology. While the tests themselves, whether of the blood or gene-based, are performed by humans, the speed, accuracy or volume with which AI can analyse and draw precise conclusions from the results is unparalleled. AI for pathology—particularly blood tests—has been the most successful because, as experts say, data on blood reports is easier given the fact that almost everyone across age groups, countries, gender and ethnicity gets blood tests done. Some products are so powerful that they now fully automate microscopy for blood and urine using AI. Among them is AI100, the flagship product of SigTuple, an Indian firm that builds AI-powered software to analyse medical imaging data. It successfully mimics the work of a qualified pathologist, and is being used by path labs such as HCG and Thyrocare across India. "The software is an AI module that analyses these images," says Dr Tathagato Rai Dastidar, SigTuple founder and CEO. "The result is then presented to the pathologist on a web browser, accessible anywhere, anytime. This not only brings down the

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"It is essential that India is not left behind in the new predictive diagnostics technology. **By tapping into our domestic manufacturing potential,** we can make it available to Indians at affordable rates"

–DR V.K. PAUL
Member, NITI Aayog

